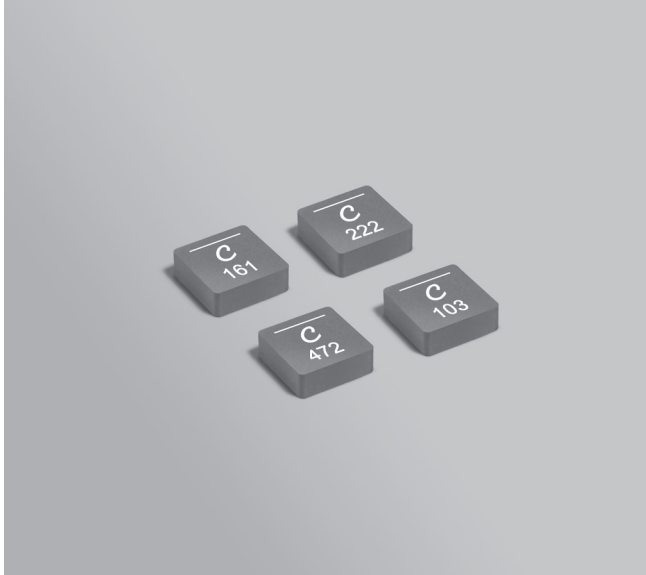




Shielded Power Inductors – XAL7030



- High current and very low DCR
- AEC-Q200 qualified
- Soft saturation makes them ideal for VRM/VRD applications.

Designer's Kit C441 contains 3 each of all values

Core material Composite

Environmental RoHS compliant, halogen free

Terminations RoHS compliant tin-silver (96.5/3.5) over copper. Other terminations available at additional cost.

Weight 0.83 – 0.86 g

Operating voltage: 60 V⁷

Ambient temperature –40°C to +125°C with (40°C rise) Irms current.

Maximum part temperature +165°C (ambient + temp rise). [Derating](#).

Storage temperature Component: –55°C to +165°C.

Tape and reel packaging: –55°C to +80°C

Resistance to soldering heat Max three 40 second reflows at +260°C, parts cooled to room temperature between cycles

Moisture Sensitivity Level (MSL) 1 (unlimited floor life at <30°C / 85% relative humidity)

Packaging 400/7" reel; 1500/13" reel Plastic tape: 16 mm wide, 0.3 mm thick, 12 mm pocket spacing, 3.3 mm pocket depth

PCB washing Tested to MIL-STD-202 Method 215 plus an additional aqueous wash. See [Doc787_PCB_Washing.pdf](#).

Part number ¹	Inductance ² ±20% (μH)	DCR (mOhms) ³		SRF typ ⁴ (MHz)	Isat ⁵ (A)	Irms (A) ⁶	
		typ	max			20°C rise	40°C rise
XAL7030-161ME_	0.16	1.15	1.26	158	60.0	24.9	32.5
XAL7030-301ME_	0.30	1.75	1.92	101	41.0	21.0	27.6
XAL7030-601ME_	0.60	3.00	3.30	72	36.0	18.0	23.0
XAL7030-102ME_	1.0	4.55	5.00	52	28.0	16.1	21.8
XAL7030-152ME_	1.5	7.60	8.36	39	23.5	11.9	15.0
XAL7030-222ME_	2.2	13.7	15.07	29	18.0	10.0	12.9
XAL7030-272ME_	2.7	15.7	17.30	32	12.8	9.2	11.4
XAL7030-332ME_	3.3	19.5	21.45	25	12.3	8.0	10.0
XAL7030-472ME_	4.7	26.1	30.00	21	10.1	6.9	9.0
XAL7030-562ME_	5.6	28.1	32.32	17	9.8	5.3	7.3
XAL7030-682ME_	6.8	45.0	51.75	15	8.7	4.4	6.8
XAL7030-822ME_	8.2	53.0	60.94	13	8.4	2.9	5.9
XAL7030-103ME_	10	60.4	69.46	12	7.7	2.6	5.3

Irms Testing

Irms testing was performed on 0.75 inch wide × 0.25 inch thick copper traces in still air.

Temperature rise is highly dependent on many factors including pcb land pattern, trace size, and proximity to other components. Therefore temperature rise should be verified in application conditions.

1. When ordering, please specify **termination** and **packaging** code:

XAL7030-103MEC

Termination: **E** = Halogen free component. RoHS compliant tin-silver over copper terminations.
Special order: **T** = RoHS tin-silver-copper (95.5/4/0.5) or **S** = non-RoHS tin-lead (63/37).

Packaging: **C** = 7" machine-ready reel. EIA-481 embossed plastic tape (400 parts per full reel). Quantities less than full reel available: in tape (not machine ready) or with leader and trailer (\$25 charge).

B = Less than full reel. In an effort to simplify our part numbering system, Coilcraft is eliminating the need for multiple packaging codes. When ordering, simply change the last letter of your part number from B to C.

D = 13" machine-ready reel. EIA-481 embossed plastic tape. Factory order only, not stocked (1500 parts per full reel).

2. Inductance tested at 1 MHz, 0.1 Vrms, 0 Adc.

3. DCR measured on a micro-ohmmeter.

4. SRF measured using Agilent/HP 4395A or equivalent.

5. DC current at 25°C that causes an inductance drop of 30% (typ) from its value without current.

[Click for temperature derating information.](#)

6. Current that causes the specified temperature rise from 25°C ambient. This information is for reference only and does not represent absolute maximum ratings. [Click for temperature derating information.](#)

7. Voltage capability varies by part number and in many cases may be higher than the listed voltage.

8. Electrical specifications at 25°C.

Refer to Doc 362 "Soldering Surface Mount Components" before soldering.



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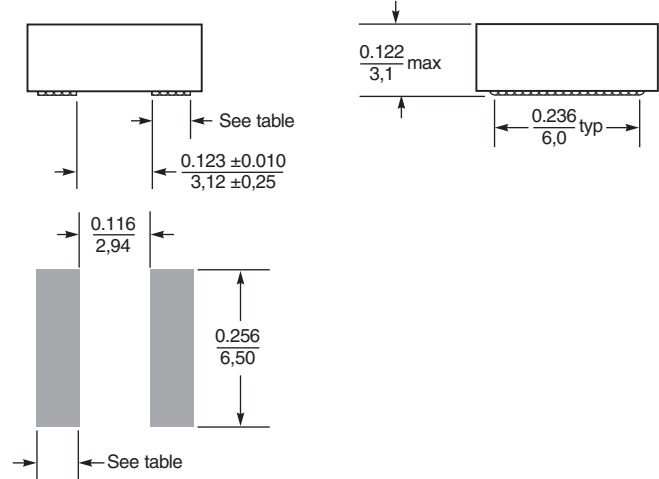
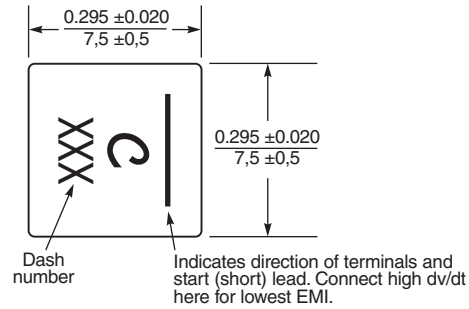
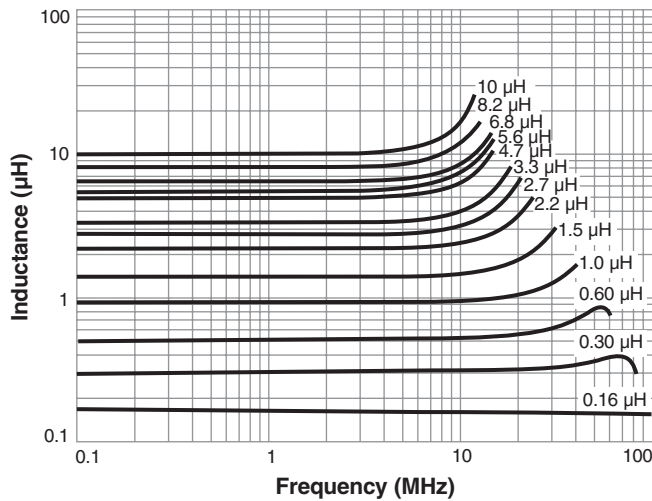
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HIGH TEMPERATURE

Shielded Power Inductors – XAL7030

Typical L vs Frequency



Recommended Land Pattern

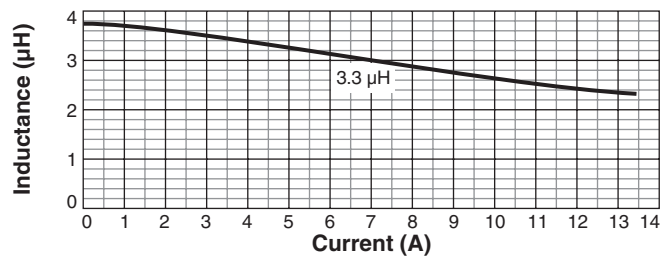
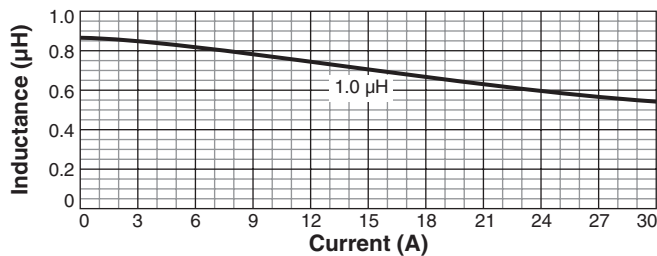
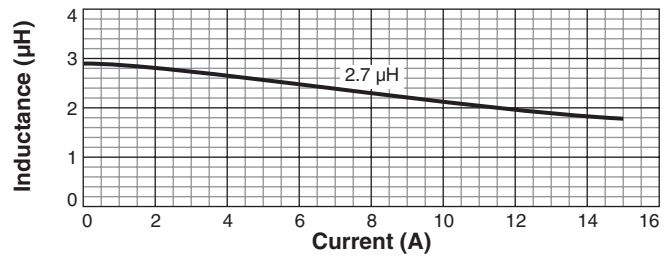
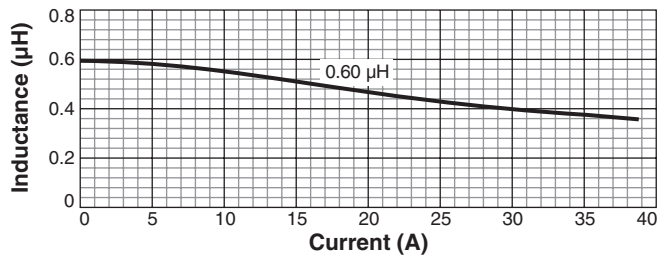
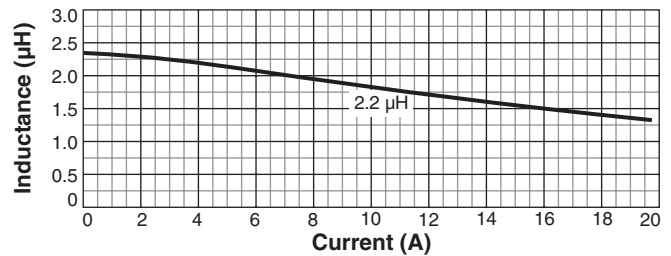
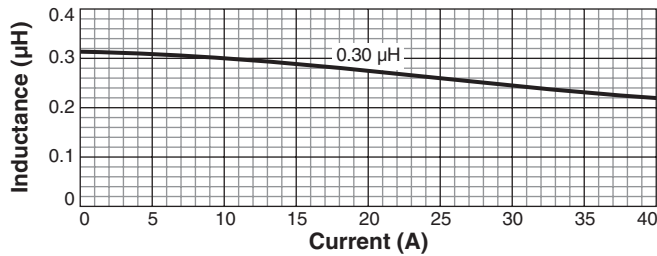
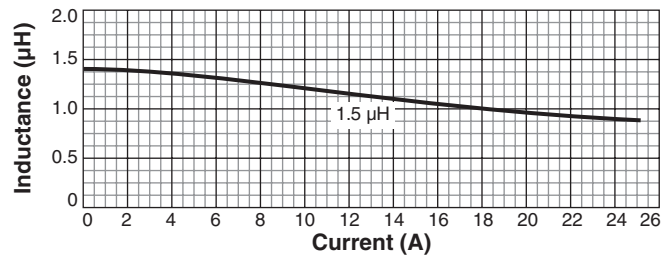
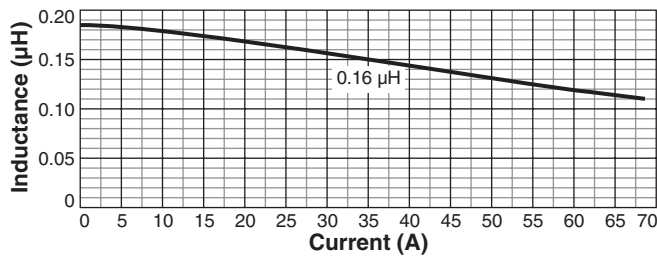
Dimensions are in $\frac{\text{inches}}{\text{mm}}$

Dash number	Terminal width $\pm 0.004 / \pm 0.10$ (in / mm)	Land pattern width (in / mm)
-161	0.063 / 1,60	0.070 / 1.78
-301	0.063 / 1,60	0.070 / 1.78
-601	0.063 / 1,60	0.070 / 1.78
-102	0.063 / 1,60	0.070 / 1.78
-152	0.063 / 1,60	0.070 / 1.78
-222	0.063 / 1,60	0.070 / 1.78
-272	0.063 / 1,60	0.070 / 1.78
-332	0.063 / 1,60	0.070 / 1.78
-472	0.055 / 1,40	0.062 / 1.58
-562	0.055 / 1,40	0.062 / 1.58
-682	0.055 / 1,40	0.062 / 1.58
-822	0.055 / 1,40	0.062 / 1.58
-103	0.055 / 1,40	0.062 / 1.58

HIGH TEMPERATURE

Shielded Power Inductors – XAL7030

L vs Current



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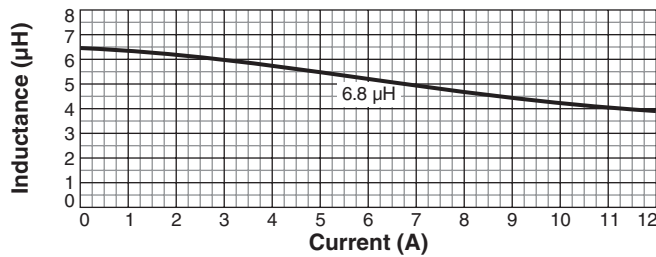
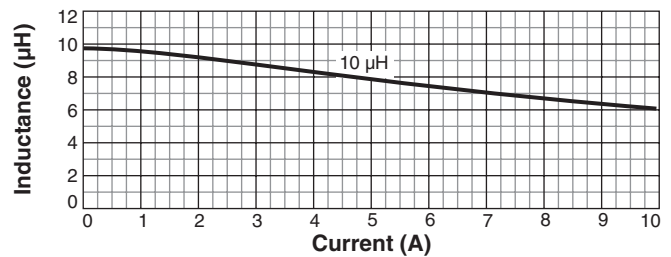
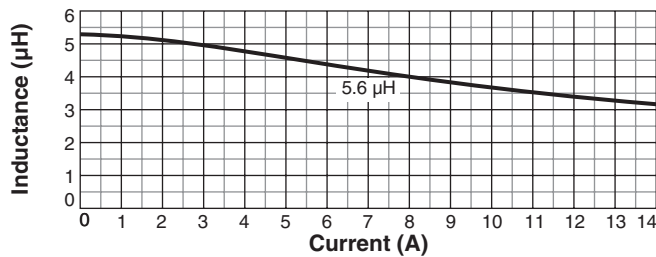
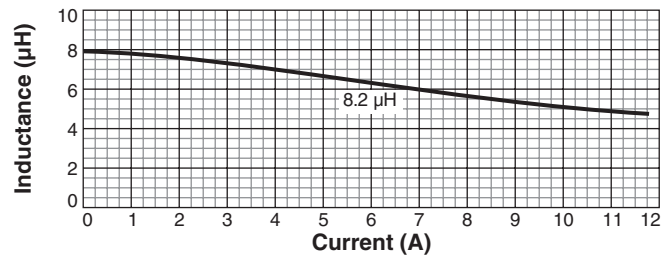
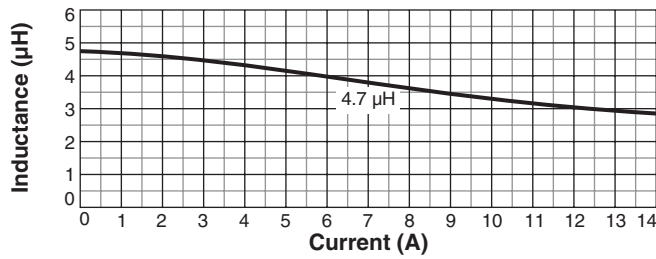
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HIGH TEMPERATURE



Shielded Power Inductors – XAL7030

L vs Current



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